Bayesian multimodeling:
Posterior and evidence estimation

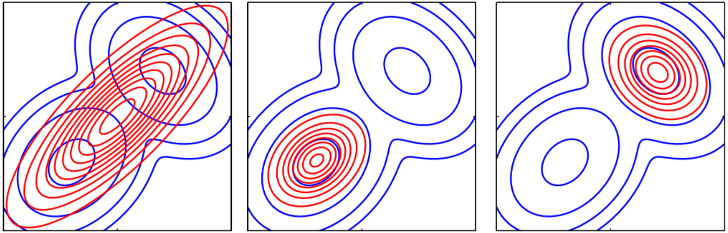
2025

Recap: ELBO

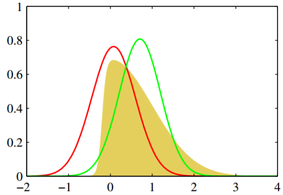
Evidence lower bound is a method of approximation of intractable distribution $p(\mathbf{w}|\mathfrak{D}, \mathbf{h})$ with a distribution $q(\mathbf{w}) \in \mathfrak{Q}$.
Evidence lower bound estimation often reduces to optimization problem

$$\log p(\mathfrak{D}|\mathbf{h}) \geq$$

$$\geq -\int_{\mathbf{w}} q(\mathbf{w}) \log \frac{p(\mathbf{w}|\mathfrak{D})}{q(\mathbf{w})} d\mathbf{w} = \mathsf{E}_{\mathbf{w}} \log p(\mathfrak{D}|\mathbf{w}) - \mathrm{KL}(q(\mathbf{w})||p(\mathbf{w}|\mathbf{h})).$$



Variational inference vs. expectation propogation (Bishop)



Laplace Approximation vs Variational inference

Recap: ELBO estimation

ELBO maximization

$$\int_{\mathbf{w}} q(\mathbf{w}) \log \frac{p(\mathbf{y}, \mathbf{w}|\mathbf{X}, \mathbf{h})}{q(\mathbf{w})} d\mathbf{w}$$

is equivalent to KL-divergence minimization between $q(\mathbf{w}) \in \mathfrak{Q}$ and posterior distribution $p(\mathbf{w}|\mathbf{y}, \mathbf{X}, \mathbf{h})$:

$$\hat{q} = \arg\max_{q \in \mathfrak{Q}} \int_{\mathbf{w}} q(\mathbf{w}) \log \frac{p(\mathbf{y}, \mathbf{w}|\mathbf{X}, \mathbf{h})}{q(\mathbf{w})} d\mathbf{w} \Leftrightarrow$$

$$\hat{q} = \arg\min_{q \in \mathfrak{Q}} \mathrm{D_{KL}}\big(q(\mathbf{w}) || p(\mathbf{w}|\mathbf{y}, \mathbf{X}, \mathbf{h})\big),$$

$$\mathrm{D_{KL}}\big(q(\mathbf{w}) || p(\mathbf{w}|\mathbf{y}, \mathbf{X}, \mathbf{h})\big) = \int_{\mathbf{w}} q(\mathbf{w}) \log \left(\frac{q(\mathbf{w})}{p(\mathbf{w}|\mathbf{y}, \mathbf{X}, \mathbf{h})} \right) d\mathbf{w}.$$

Outline, global variational methods

- Can we use something except Gaussian distribution?
  - Yes, we can
- Does it need to have an analytical form?
  - No
- Does it need to have some specific properties except continuity?
  - No
- Do we need a parametric distribution to approximate posterior?
  - No
- Is it always about Evidence approximation?
  - In general, no. We can use other probability distances.

Reparametrization trick: why do we need it [1]

Suppose we have a function $f(\mathbf{w})$ depending on a stochastic variable $\mathbf{w} \sim q_{\boldsymbol{\theta}}(\mathbf{w})$.
How we can calculate an gradient of expectation:

$$\nabla_{\boldsymbol{\theta}} \mathrm{E}_q f(\mathbf{w})?$$

Naive way:

$$\nabla_{\boldsymbol{\theta}} \mathrm{E}_q f(\mathbf{w}) = \int_{\mathbf{w}} \nabla_{\boldsymbol{\theta}} \left(f(\mathbf{w}) q(\mathbf{w}) d\mathbf{w} \right) =$$

$$= \int_{\mathbf{w}} q(\mathbf{w}) \nabla_{\boldsymbol{\theta}} f(\mathbf{w}) d\mathbf{w} + \int_{\mathbf{w}} f(\mathbf{w}) \nabla_{\boldsymbol{\theta}} q(\mathbf{w}) d\mathbf{w} =$$

$$= \mathrm{E}_{\mathbf{w}} \nabla_{\boldsymbol{\theta}} f(\mathbf{w}) + \int_{\mathbf{w}} f(\mathbf{w}) \nabla_{\boldsymbol{\theta}} q(\mathbf{w}) d\mathbf{w}.$$

[1] https://gregorygundersen.com/blog/2018/04/29/reparameterization/

Reparametrization trick

Reparamterization idea:

$$\varepsilon = S_{\boldsymbol{\theta}}(\mathbf{w}), \quad \mathbf{w} = S_{\boldsymbol{\theta}}^{-1}(\varepsilon).$$

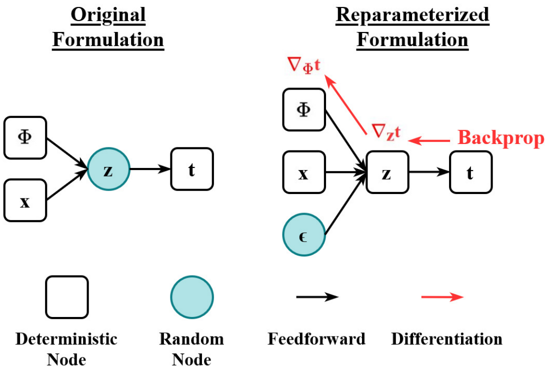
Then:

$$\nabla_{\boldsymbol{\theta}} E_q f(\mathbf{w}) = \mathrm{E}_{\varepsilon} \nabla_{\boldsymbol{\theta}} f(S_{\boldsymbol{\theta}}^{-1}(\varepsilon)).$$

Example:

$$w \sim \mathcal{N}(\mu, \sigma^2) \to S(w) = \frac{w - \mu}{\sigma} \sim \mathcal{N}(0, 1).$$

Challenge: calculation of S^{-1} is an expensive operation.



Source: wikipedia

Normalizing Flows

Given an invertible smooth mapping $\mathbf{g}$ (flow) and a distribution $\mathbf{z} \sim q$.
Then $q(\mathbf{g}(\mathbf{z}))$ is a distribution:

$$\mathbf{g}(g(\mathbf{z})) = q(\mathbf{z}) \left(\det \frac{\partial g}{\partial \mathbf{z}} \right)^{-1}.$$

Example: planar flow:

$$\mathbf{g}(\mathbf{z}) = \mathbf{z} + \mathbf{w}_1 \boldsymbol{\sigma}(\mathbf{w}_2{}^\mathsf{T} \mathbf{x}).$$

Implicit reparametrization trick

$$\nabla_{\boldsymbol{\theta}} E_q f(\mathbf{w}) = \mathsf{E}_q \nabla_{\mathbf{w}} f(\mathbf{w}) \nabla_{\boldsymbol{\theta}} \mathbf{w}.$$

Use a total gradient formula for $\varepsilon = S_{\boldsymbol{\theta}}(\mathbf{w})$:

$$\nabla_{\mathbf{w}} S_{\boldsymbol{\theta}}(\mathbf{w}) \nabla_{\boldsymbol{\theta}} \mathbf{w} + \nabla_{\boldsymbol{\theta}} S_{\boldsymbol{\theta}}(\mathbf{w}) = 0 \to$$

$$\to \nabla_{\boldsymbol{\theta}} \mathbf{w} = -(\nabla_{\mathbf{w}} S_{\boldsymbol{\theta}}(\mathbf{w}))^{-1} \nabla_{\boldsymbol{\theta}} S_{\boldsymbol{\theta}}.$$

Obtain an expression without inverse function for S.
For 1d samples we can use, for example:

$$S(\mathbf{w}) = F(\mathbf{w}|\boldsymbol{\theta}) \sim \mathcal{U}(0, 1).$$

Table 4: Test negative log-likelihood (lower is better) for VAE on MNIST. Mean $\pm$ standard deviation over 5 runs. The von Mises-Fisher results are from [9].

Prior	Variational posterior	$D = 2$	$D = 5$	$D = 10$	$D = 20$	$D = 40$
$\mathcal{N}(0, 1)$	$\mathcal{N}(\mu, \sigma^2)$	131.1 ± 0.6	107.9 ± 0.4	92.5 ± 0.2	88.1 ± 0.2	88.1 ± 0.0
$\mathrm{Gamma}(0.3, 0.3)$	$\mathrm{Gamma}(\alpha, \beta)$	132.4 ± 0.3	108.0 ± 0.3	94.0 ± 0.3	90.3 ± 0.2	90.6 ± 0.2
$\mathrm{Gamma}(10, 10)$	$\mathrm{Gamma}(\alpha, \beta)$	135.0 ± 0.2	107.0 ± 0.2	92.3 ± 0.2	88.3 ± 0.2	88.3 ± 0.1
$\mathrm{Uniform}(0, 1)$	$\mathrm{Beta}(\alpha, \beta)$	128.3 ± 0.2	107.4 ± 0.2	94.1 ± 0.1	88.9 ± 0.1	88.6 ± 0.1
$\mathrm{Beta}(10, 10)$	$\mathrm{Beta}(\alpha, \beta)$	131.1 ± 0.4	$\mathbf{106.7} \pm 0.1$	$\mathbf{92.1} \pm 0.2$	$\mathbf{87.8} \pm 0.1$	$\mathbf{87.7} \pm 0.1$
$\mathrm{Uniform}(-\pi, \pi)$	$\mathrm{vonMises}(\mu, \kappa)$	$\mathbf{127.6} \pm 0.4$	107.5 ± 0.4	94.4 ± 0.5	90.9 ± 0.1	91.5 ± 0.4
$\mathrm{vonMises}(0, 10)$	$\mathrm{vonMises}(\mu, \kappa)$	130.7 ± 0.8	107.5 ± 0.5	92.3 ± 0.2	$\mathbf{87.8} \pm 0.2$	87.9 ± 0.3
$\mathrm{Uniform}(S^D)$	$\mathrm{vonMisesFisher}(\boldsymbol{\mu}, \kappa)$	132.5 ± 0.7	108.4 ± 0.1	93.2 ± 0.1	89.0 ± 0.3	90.9 ± 0.3

Gradient descent for evidence estimation, Maclaurin et. al, 2015

Consider posterior probability maximization:

$$L = -\log\, p(\mathfrak{D}, \mathbf{w}|\mathbf{h}) = -\sum_{\mathfrak{D} \in \mathfrak{D}} \log\, p(\mathfrak{D}|\mathbf{w}, \mathbf{h}) p(\mathbf{w}|\mathbf{h})$$

Optimize neural network in a multi-start regime with r initial parameter values $\mathbf{w}_1, \dots, \mathbf{w}_r$ using (stochastic) gradient descent:

$$\mathbf{w}' = T(\mathbf{w}).$$

The parameter vectors $\mathbf{w}_1, \dots, \mathbf{w}_r$ are from some latent distribution $q(\mathbf{w})$.

Entropy

We can rewrite variational inference using differential entropy term:

$$\log\, p(\mathfrak{D}|\mathbf{f}) \geq \int_{\mathbf{w}} q(\mathbf{w}) \log\, \frac{p(\mathfrak{D}, \mathbf{w}|\mathbf{h})}{q(\mathbf{w})} d\mathbf{w} =$$

$$\mathsf{E}_{q(\mathbf{w})}[\log\, p(\mathfrak{D}, \mathbf{w}|\mathbf{h})] + \mathsf{S}(q(\mathbf{w})),$$

where $\mathsf{S}(q(\mathbf{w}))$ is a differential entropy:

$$\mathsf{S}(q(\mathbf{w})) = -\int_{\mathbf{w}} q(\mathbf{w}) \log\, q(\mathbf{w}) d\mathbf{w}.$$

Gradient descent for evidence estimation

Statement

Let L be a Lipschitz function, and optimization operator be a bijection. Then entropy difference for two steps is:

$$\mathsf{S}(q'(\mathbf{w})) - \mathsf{S}(q(\mathbf{w})) \simeq \frac{1}{r}\sum_{g=1}^{r}\Big(-\beta \mathrm{Tr}[\mathbf{H}(\mathbf{w}'^{g})] - \beta^2 \mathrm{Tr}[\mathbf{H}(\mathbf{w}'^{g})\mathbf{H}(\mathbf{w}'^{g})]\Big).$$

Final estimation for the τ optimization step:

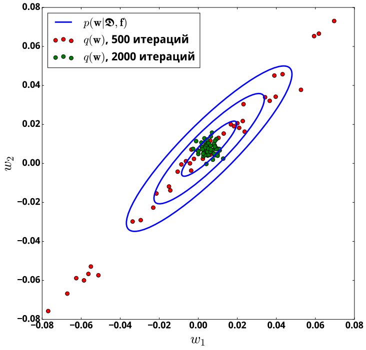
$$\log \hat{p}(\mathbf{Y}|\mathfrak{D}, \mathbf{h}) \sim \frac{1}{r}\sum_{g=1}^{r} L(\mathbf{w}_{\tau}^{g}, \mathfrak{D}, \mathbf{Y}) + \mathsf{S}(q^0(\mathbf{w})) +$$

$$+\frac{1}{r}\sum_{b=1}^{\tau}\sum_{g=1}^{r}\Big(-\beta \mathrm{Tr}[\mathbf{H}(\mathbf{w}_b^g)] - \beta^2 \mathrm{Tr}[\mathbf{H}(\mathbf{w}_b^g)\mathbf{H}(\mathbf{w}_b^g)]\Big),$$

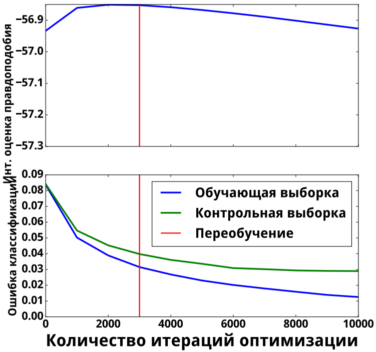
$\mathbf{w}_b^g$ is a parameter vector for optimization g on the step b, $\mathsf{S}(q^0(\mathbf{w}))$ is an initial entropy.

Overfitting, Maclaurin et. al, 2015

Gradient descent does not optimize KL-divergence $\mathrm{KL}(q(\mathbf{w})||p(\mathbf{w}|\mathfrak{D}, \mathbf{h}))$. Evidence estimation gets worse while optimization tends to the optimal parameter values. This can be considered as a overfitting start.



Convergence



Overfitting start

Stochastic gradient Langevin dynamics

A modification of SGD:

$$T = \mathbf{w} - \beta \nabla L + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \frac{\beta}{2})$$

where β changes with a number of iterations:

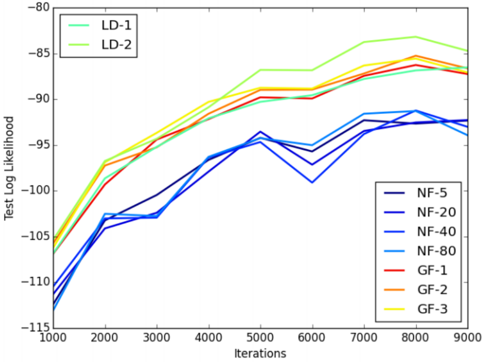
$$\sum_{\tau=1}^{\infty} \beta_\tau = \infty, \quad \sum_{\tau=1}^{\infty} \beta_\tau^2 < \infty.$$

Statement [Welling, 2011]. Distribution $q^\tau(\mathbf{w})$ converges to posterior distribution $p(\mathbf{w}|\mathbf{X}, \mathbf{f})$.
Entropy adjustment:

$$\hat{\mathsf{S}}(q^\tau(\mathbf{w})) \geq \frac{1}{2}|\mathbf{w}|\log\left(\exp\left(\frac{2\mathsf{S}(q^\tau(\mathbf{w}))}{|\mathbf{w}|}\right) + \exp\left(\frac{2\mathsf{S}(\epsilon)}{|\mathbf{w}|}\right)\right).$$

Stochastic gradient Langevin dynamics for generative models

Altieri et al., 2015: sample latent variable $\mathbf{z}$ and use SGLD as a normalizing flow.



Stein operator

Given a smooth probability function p and a smooth vector function ϕ. Define a Stein operator as the following:

$$\mathcal{A}_p\boldsymbol{\phi}(\mathbf{x}) = \nabla_{\mathbf{x}} \log p(\mathbf{x})\boldsymbol{\phi}^{\mathsf{T}} + \nabla_{\boldsymbol{\phi}}\boldsymbol{\phi}(\mathbf{x}).$$

Stein's identity:

$$\mathsf{E}_{\mathbf{x}\sim p}\mathcal{A}_p\boldsymbol{\phi}(\mathbf{x}) = 0.$$

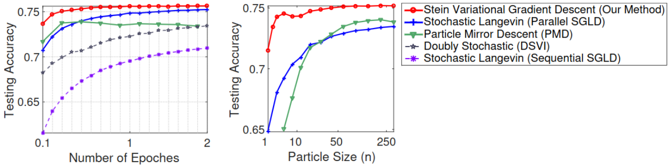
If we use q instead of p in the $\mathcal{A}_p$ we get a non-zero result, but close to zero as soon as p is close to q.
Let $T(\mathbf{x}) = \mathbf{x} + \varepsilon\boldsymbol{\phi}(\mathbf{x})$. Then:

$$\nabla_{\varepsilon} KL(q||p)|_{\varepsilon=0} = \mathsf{E}_{\mathbf{x}\sim q}\mathrm{trace}\mathcal{A}_p\boldsymbol{\phi}.$$

Given a kernel $\mathbf{K}$, the optimal ϕ for minimizing KL is:

$$\phi^*(\mathbf{x}') = \mathsf{E}_{\mathbf{x}\sim q}\nabla_{\mathbf{x}} \log\ p(\mathbf{x})\mathbf{K}(\mathbf{x},\mathbf{x}') + \nabla_{\mathbf{x}}\mathbf{K}(\mathbf{x},\mathbf{x}').$$

Stein operator: results



(a) Particle size $n = 100$ (b) Results at 3000 iteration (≈ 0.32 epoches)

	Avg. Test RMSE		Avg. Test LL		Avg. Time (Secs)	
Dataset	**PBP**	**Our Method**	**PBP**	**Our Method**	**PBP**	**Ours**
Boston	2.977 ± 0.093	$\mathbf{2.957 \pm 0.099}$	-2.579 ± 0.052	$\mathbf{-2.504 \pm 0.029}$	18	**16**
Concrete	5.506 ± 0.103	$\mathbf{5.324 \pm 0.104}$	-3.137 ± 0.021	$\mathbf{-3.082 \pm 0.018}$	33	**24**
Energy	1.734 ± 0.051	$\mathbf{1.374 \pm 0.045}$	-1.981 ± 0.028	$\mathbf{-1.767 \pm 0.024}$	25	**21**
Kin8nm	0.098 ± 0.001	$\mathbf{0.090 \pm 0.001}$	0.901 ± 0.010	$\mathbf{0.984 \pm 0.008}$	118	**41**
Naval	0.006 ± 0.000	$\mathbf{0.004 \pm 0.000}$	3.735 ± 0.004	$\mathbf{4.089 \pm 0.012}$	173	**49**
Combined	4.052 ± 0.031	$\mathbf{4.033 \pm 0.033}$	-2.819 ± 0.008	$\mathbf{-2.815 \pm 0.008}$	136	**51**
Protein	4.623 ± 0.009	$\mathbf{4.606 \pm 0.013}$	-2.950 ± 0.002	$\mathbf{-2.947 \pm 0.003}$	682	**68**
Wine	0.614 ± 0.008	$\mathbf{0.609 \pm 0.010}$	-0.931 ± 0.014	$\mathbf{-0.925 \pm 0.014}$	26	**22**
Yacht	$\mathbf{0.778 \pm 0.042}$	0.864 ± 0.052	$\mathbf{-1.211 \pm 0.044}$	-1.225 ± 0.042	25	25
Year	$8.733 \pm \text{NA}$	$\mathbf{8.684 \pm NA}$	$-3.586 \pm \text{NA}$	$\mathbf{-3.580 \pm NA}$	7777	**684**

Rényi Divergence Variational Inference

Rényi's α-divergence

$$D_\alpha[p||q] = \frac{1}{\alpha - 1} \log \int p(\boldsymbol{\theta})^\alpha q(\boldsymbol{\theta})^{1-\alpha} d\boldsymbol{\theta} \tag{1}$$

1. continuous and non-decreasing on $\alpha \in \{\alpha : |D_\alpha| < +\infty\}$
2. for $\alpha \notin \{0, 1\}, D_\alpha[p||q] = \frac{\alpha}{1-\alpha} D_{1-\alpha}[p||q] \rightarrow D_\alpha[p||q] \leq 0, \alpha < 0$

See also 2 talks: from Kseniia Petrushina and Galina Boeva, 2023.

Different divergence functions

α	Definition	Notes
$\alpha \to 1$	$\int p(\boldsymbol{\theta}) \log \frac{p(\boldsymbol{\theta})}{q(\boldsymbol{\theta})} d\boldsymbol{\theta}$	*Kullback-Leibler (KL) divergence*, used in VI ($\mathrm{KL}[q\|\|p]$) and EP ($\mathrm{KL}[p\|\|q]$)
$\alpha = 0.5$	$-2\log(1 - \mathrm{Hel}^2[p\|\|q])$	function of the square *Hellinger distance*
$\alpha \to 0$	$-\log \int_{p(\boldsymbol{\theta})>0} q(\boldsymbol{\theta}) d\boldsymbol{\theta}$	zero when $\mathrm{supp}(q) \subseteq \mathrm{supp}(p)$ (not a divergence)
$\alpha = 2$	$-\log(1 - \chi^2[p\|\|q])$	proportional to the χ^2-divergence
$\alpha \to +\infty$	$\log \max_{\boldsymbol{\theta} \in \Theta} \frac{p(\boldsymbol{\theta})}{q(\boldsymbol{\theta})}$	*worst-case regret* in *minimum description length principle* [24]

- When $\alpha \to 0$, we get an approximate Evidence (like in IWAE).
- When $\alpha \to 1$, we get ELBO.
- When $\alpha \to \infty$ we get mode-seeking (also called zero-forcing) optimization.
- When $\alpha \to -\infty$ we get mass-preserving optimization.

Sampling: Naive method

$$I = \mathsf{E}f = \int_w f(\mathbf{w})p(\mathbf{w})d\mathbf{w}.$$

Approximate:

$$\hat{I} = \frac{1}{N}\sum_{\mathbf{w}\sim p(\mathbf{w})} f(\mathbf{w}).$$

Why this does not work?

Properties

Integral estimation:

- strongly consistent : $\hat{I} \rightarrow^{\text{a.s.}} I$
- Unbiased: $\mathrm{E}\hat{I} = I$
- Assymptotically normal;
- $\mathrm{D}\hat{I} = O(\frac{1}{N})$.
- **Challenge:** we need to sample from p.

Inverse transform sampling

Let T be a invertible function from $u \sim \mathcal{U}(0,1)$ to some random variable distribution $p(w)$. Then

$$F_w(t) = p(w \leq t) = p(T(u) \leq t) = p(u \leq T^{-1}(t)) = T^{-1}(t).$$

Therefore $F_u^{-1} = T$.

Example

$$w = \lambda \text{exp}(-\lambda t).$$

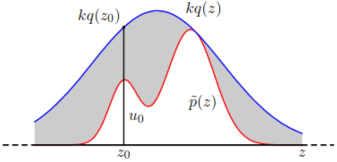
$$F_w(t) = 1 - \text{exp}(-\lambda t).$$

$$F_w^{-1}(u) = -1\frac{1}{\lambda}\log(1-u).$$

Rejection sampling

- Given $p(w)$ (up to normalizing constant)
- Set distribution q
- Set value k so that $kq(w) \geq p(z)$ for all z
- In a loop:
  - Sample $w_0 \sim q$
  - Sample $u \sim \mathcal{U}(0, kq(w_0))$
  - If $u \leq p(w_0)$, use it as a sample from $p(w)$

Core idea: samples u are uniform in a region limited by $p(w)$.



Bishop, 2006

Importance sampling

Consider the case when we cannot sample from $p(w)$, but we can estimate likelihood and want to estimate the integral

$$\mathsf{E}f = \int f(w)p(w)dz.$$

Let q be an auxilary distribution:

$$\mathsf{E}f = \int f(w)p(w)dw = \int f(w)\frac{p(w)}{q(w)}q(w)dw \approx \frac{1}{L}\sum_{l=1}^{L}\frac{p(w^l)}{q(w^l)}f(w^l).$$

MCMC

Basic idea: Sample similar to rejection sampling, but q is a Markov distribution with conditioning on the previous step.
We want the stationary (limiting) distribution to be equal to our $p(w)$.
Sufficient condition

$$p(w')T(w|w') = p(w)T(w'|w).$$

Metropolis-Hastings algorithm

- Sample new $w' \sim q(w|w^t)$.
- Accept with probability $A(w'|w^t) = \min\left(1, \frac{p(w')q(w^t|w')}{p(w^t)q(w'|w^t)}\right)$.
- If accepted: $w^{t+1} = w'$,
- Otherwise: $w^{t+1} = w^t$.

Sufficient condition is satisfied:

$$p(w')T(w|w') = p(w)T(w'|w) = p(w')T(w'|w^t) = p(w')q(w'|w^t)A(w'|w^t) =$$

$$= p(w^t)q(w^t|w')A(w^t|w').$$

- Samples are correlated. We can decorrelate sample using each k sample.
- Works better in high-dimmensional settings than rejection sampling.
- Good choice of q is the main challenge for the algorithm.

MCMC and variational inference

MCMC idea: Sample from the simple distribution and accpet them, if the ratio is greater than some threshold:

$$\min \left(1, \frac{p(\mathbf{w}^\tau|\mathbf{y}, \mathbf{X}, \mathbf{h})}{p(\mathbf{w}^{\tau-1}|\mathbf{y}, \mathbf{X}, \mathbf{h})}\right),$$

where $\mathbf{w}^\tau$ is set based on the previous sample:

$$\mathbf{w}^\tau = T(\mathbf{w}^{\tau-1}).$$

Salimans et al., 2014: let's interperete the sequence of some operator T application as a variational optimization:

$$T^1 \circ \dots T^\eta(\mathbf{w}) \to p(\mathbf{w}^\tau|\mathbf{y}, \mathbf{X}, \mathbf{h}).$$

Maclaurin et. al, 2015: use gradient descent as such operator. Do not reject samples at all.

Autoencoder: generative model?

(Alain, Bengio 2012): consider regularized autoencoder:

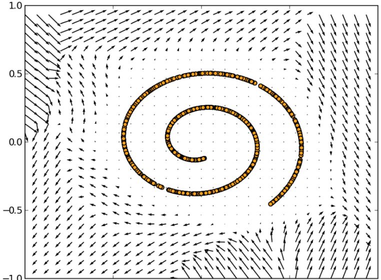
$$||\mathbf{f}(\mathbf{x}, \sigma) - \mathbf{x}||^2,$$

where σ is a noise level.

Then

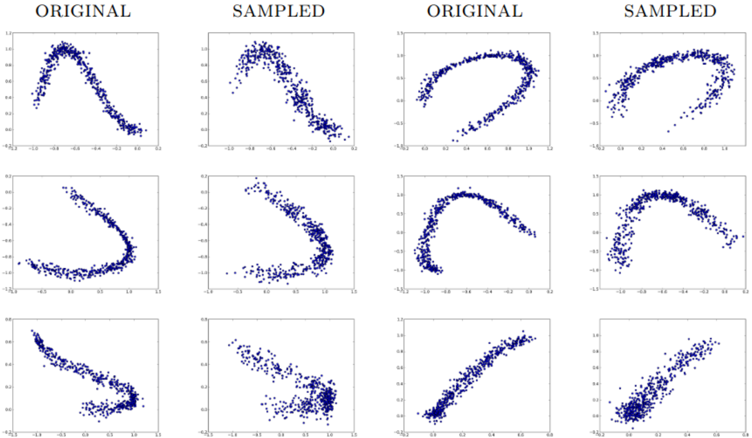
$$\frac{\partial \log p(x)}{\partial x} = \frac{||\mathbf{f}(\mathbf{x}, \sigma) - \mathbf{x}||^2}{\sigma^2} + o(1) \text{ when } \sigma \to 0.$$

Vector field induced by reconstruction error



Autoencoder for sampling

$$A = \frac{p(x^*)}{p(x)} = \exp\left(E(x) - E(x^*)\right) \approx \frac{\partial E(x)}{\partial x}^{\mathsf{T}}(x^* - x) + o(||x - x^*||).$$



What Are Bayesian Neural Network Posteriors Really Like?

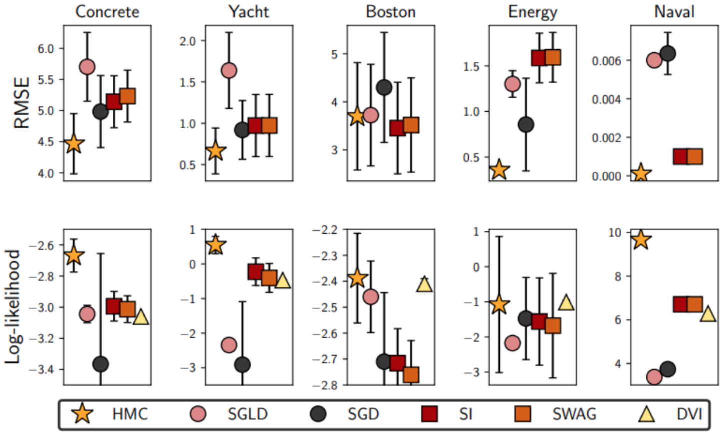
Izmailov et al., 2021:[2]

- HMC for posterior distribuition estimation for deep models on some standard datasets.
- Resources: 512 TPU

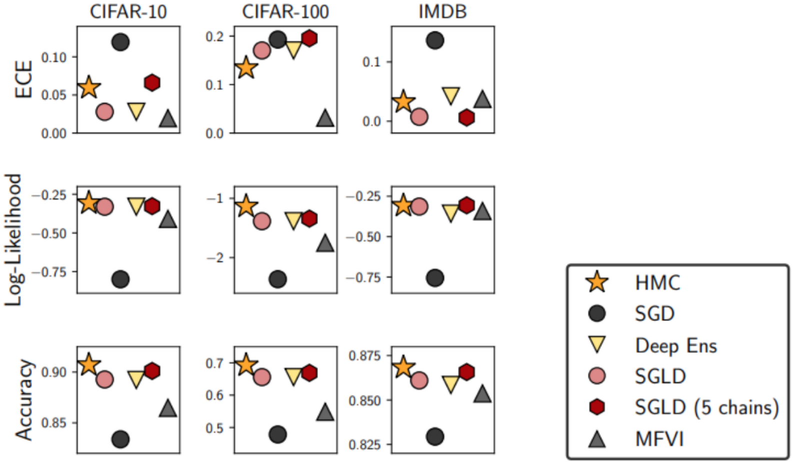
[2] https://docs.google.com/presentation/d/1WPjqKw3b-TpPSaHcwhqFsuAE575_nDigt5SVoP3CoNI/edit?usp=sharing

What Are Bayesian Neural Network Posteriors Really Like?

BNN evaluation: UCI

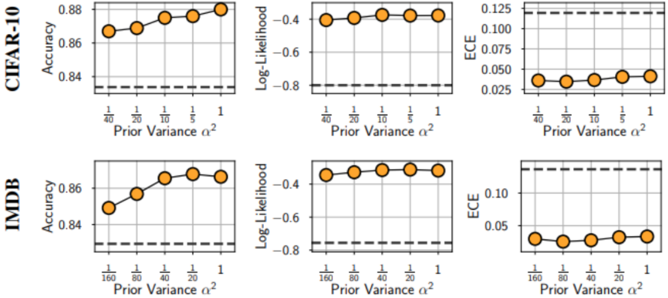


BNN evaluation: CIFAR and IMDB



What Are Bayesian Neural Network Posteriors Really Like?

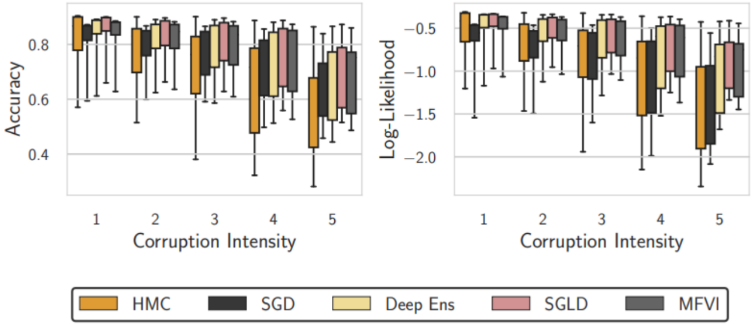
Effect of priors



HMC BNNs are fairly robust to Gaussian prior variance.

What Are Bayesian Neural Network Posteriors Really Like?

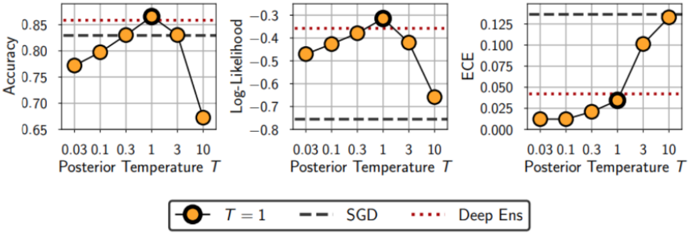
Train on CIFAR-10, test on CIFAR-10-C



What Are Bayesian Neural Network Posteriors Really Like?

Posterior temperature effect

- We have already seen that BNNs can do well at T=1
- What is the effect of T then?



Cold posteriors are not required for good results and in fact can hurt performance!

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